Homework 3: Qualitative Analysis II — The Chemistry

Paper track. Pairs with Home Lab 3. Do this on your own.

Part A: Formula recall (1 pt each)

Write the formula from memory:

1. Baking soda 2. Washing soda 3. Calcium carbonate 4. Sucrose
2. Starch 6. Table salt 7. Gypsum 8. Epsom salt 9. Citric acid
3. Vitamin C 11. Sodium acetate 12. Sand

Part B: Why the tests work (short answer)

1. **(4 pts)** Explain why and what makes corn starch turn black in iodine. *(Aim for four scoring points: the helix, the iodine going inside, the charge-transfer complex, the light absorption / color.)*
2. **(2 pts)** Is the iodine–starch test a chemical reaction? What is the correct term for what happens?
3. **(2 pts)** Amylose gives a blue-black color; amylopectin gives red-brown. Why the difference?
4. **(2 pts)** A powder turns iodine **colorless**. What is it, and why does that happen? How is this different from the starch result?
5. **(2 pts)** Why does chalk fizz more slowly with acid than baking soda does, even though both are carbonates?

Part C: Distinguish (2 pts each)

For each pair, name the single best test to tell them apart and the result for each:

1. Baking soda vs. baking powder
2. Table salt vs. citric acid
3. Corn starch vs. calcium carbonate
4. Granulated sugar vs. table salt

Part D: Benedict's (2 pts)

A question says "Benedict's solution turned brick-red." What does that mean? Would you expect table sugar to do this?

Answer Key

Part A: 1. $NaHCO_3$ 2. Na_2CO_3 3. $CaCO_3$ 4. $C_{12}H_{22}O_{11}$ 5. $(C_6H_{10}O_5)_n$ 6. NaCl 7. $CaSO_4{\cdot}2H_2O$ 8. $MgSO_4{\cdot}7H_2O$ 9. $C_6H_8O_7$ 10. $C_6H_8O_6$ 11. $NaC_2H_3O_2$ 12. SiO_2

Part B:

1. Starch contains **amylose**, whose molecules coil into a **helix** (1); **iodine, as triiodide ions, slides inside** the helix (1); the trapped iodine and starch form a **charge-transfer complex** (1) that **absorbs visible light**, making it look blue-black (1).
2. No — it's an **inclusion complex** (a clathrate). The iodine is physically trapped, not chemically changed.
3. Amylose is a long straight chain that can hold a long chain of iodine; amylopectin is branched, and its short branches can't fit a long iodine chain, so it only weakly complexes → red-brown.
4. Vitamin C (ascorbic acid) — it is a **reducing agent** and chemically converts the iodine to colorless iodide, so there's no iodine left to give a color. Unlike the starch test, this **is** a real chemical reaction.
5. Chalk (calcium carbonate) is **insoluble**, so the acid can only reach the surface of each grain; baking soda dissolves, so the whole amount reacts at once.

Part C:

1. **Iodine** — baking powder turns blue-black (it contains corn starch); baking soda doesn't.
2. **pH** — citric acid is acidic (pH ~2); salt is neutral (pH 7). (Or taste-safe note: never taste unknowns.)
3. **Water solubility** — corn starch stays a cloudy suspension / "oobleck"; calcium carbonate also stays cloudy but **fizzes with acid** and corn starch doesn't, and corn starch is iodine-positive. Best single test: iodine (starch +) or acid (carbonate fizzes).
4. **Heat** — sugar melts and caramelizes; salt crackles and does not melt. (Also: crystal shape — salt is cubic.)

Part D: A reducing sugar (glucose, fructose, maltose, lactose) is present. No — table sugar (sucrose) is a non-reducing sugar and stays blue unless it is first broken down.